DESN2000 SENG

Preliminary Report

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Contents

1 Introduction and Report Scope

The previous report used surveys and interviews with four stakeholder groups to define the problem and develop personas. This report applies the Ideate stage to that evidence. It compares three technical concepts, selects one through a weighted decision matrix, and translates the selected direction into a Jira backlog and a preliminary prototype. The report focuses on the student journey from accommodation search to an agreed shared living arrangement. Property manager stories remain exploratory and are assigned to the product backlog.

2 Problem Statement and Research Justification

2.1 Problem Statement

The previous stage established the following problem statement.

Create a mobile platform which allows university students and shared accommodation residents to seamlessly manage household responsibilities, expenses and housemate communication, while also enabling property managers to streamline tenancy administration and tenant communication, to reduce conflict between tenants, streamline property management, and improve the overall shared living experience.

The statement covers student and property manager needs. This report narrows the preliminary prototype to the student path. A student searches for an option, checks verification, compares compatibility, starts contact in the app, and acknowledges a household rule. Property manager functions remain outside Sprint 1.

2.2 Research Justification

Table 1 links the strongest Part 1 evidence to a user need and a backlog response. This traceability keeps the proposed features grounded in the research.

Table 1. Research traceability.

Research evidence	User need	Backlog response
All 16 international respondents reported concern about rental scams. A further 43.75% had encountered misleading or false listing information.	The international student persona needs clear evidence about a profile or listing before making contact.	Show profile verification status in US-11 and clear listing details in US-13.
87.5% rated housemate compatibility as very or extremely important. A further 68.75% had experienced conflict with a previous housemate.	The long term housemate persona needs to identify practical mismatches before committing.	Compare living preferences in US-12.
44% identified communication difficulty as the hardest part of the search. More than 80% were willing to use preference based matching.	The international student persona needs first contact without early disclosure of personal details.	Keep first contact inside the app through US-09.
All surveyed domestic students relied on verbal household agreements and had experienced conflict over responsibilities. A further 66.67% wanted better chore tracking.	A new share house resident needs accepted expectations to remain visible after move in.	Store acknowledged household rules through US-02 and US-10.

3 Solution Ideation

3.1 Research to Ideation Transition

The evidence in Table 1 was reframed as four design needs. Table 2 sets the boundary for concept generation and later evaluation.

Table 2. Research issues and design needs.

Research issue	Design need
Scam concern and misleading information	Visible verification status and clear listing data
Conflict caused by poor living fit	Structured preference comparison
Risk during first contact	In app messaging before contact disclosure
Verbal household expectations	Recorded rules with acknowledgement

The resulting design question was as follows.

> *How might we help students move from an uncertain accommodation search to an informed shared living agreement?*

Each concept was assessed against this question.

3.2 Generated Technical Concepts

The team generated three concepts with different technical boundaries. Table 3 gives the comparison. Tables 4 to 6 describe how each concept operates.

Table 3. Overview of the generated concepts.

Concept	Name	Main user task	Technical boundary
A	Verified Contact Gateway	Check status before first contact	Verification state and protected messaging
B	Explainable Compatibility Matcher	Compare practical living fit	Preference model and explainable score
C	Shared Living Agreement	Make a shared expectation explicit	Agreement proposal and acknowledgement state

Table 4. Concept A. Verified Contact Gateway.

Aspect	Description
Purpose	Reduce uncertainty during first contact with an unfamiliar person.
Operation	The system records each verification check as *unverified*, *pending*, *verified*, *rejected* or *expired*. The interface shows the status and hides submitted evidence. Messaging does not expose personal contact details.
Research fit	Responds to scam concern and reluctance to contact unknown people.
Limitation	A verified identity does not establish compatibility or future conduct.

Table 5. Concept B. Explainable Compatibility Matcher.

Aspect	Description
Purpose	Help users identify practical mismatches before committing.
Operation	Users record living preferences and mark mandatory constraints. The matcher removes candidates who breach a mandatory constraint. It then calculates a weighted score and shows the factors that affected the result.
Research fit	Responds to the importance of daily living compatibility in the Part 1 findings.
Limitation	The score supports discussion but cannot predict a successful household.

Table 6. Concept C. Shared Living Agreement.

Aspect	Description
Purpose	Make a limited household expectation explicit before shared living begins.
Operation	A user proposes one rule. The rule remains pending until the required household members acknowledge it. Full chore and expense management remains outside this concept.
Research fit	Responds to the domestic evidence about verbal arrangements and responsibility conflict.
Limitation	It begins after a household has formed and does not address trust or compatibility alone.

Each concept isolates a different technical mechanism. Concept A addresses trust, Concept B addresses compatibility and Concept C addresses agreement state. All three remained candidates for the weighted evaluation.

3.3 Ideation Method

The team moved from research to concepts by expanding an early mind map and then bundling related responses. This documented sequence aligns with Stanford d.school's Idea Expedition, which begins with How Might We questions, expands the possible responses, identifies promising ideas and develops them into concepts [1]. Its guidance treats the question as a focused prompt that can still lead to several relevant answers [2]. The design question in Section 3.1 served this role.

The team first used a mind map to connect user difficulties with ways the platform might respond. It then kept each recorded response separate long enough to compare how the system might act. In the final pass, related responses were bundled into concept families. This follows the IDEO.org method of grouping related ideas and combining their strongest parts into fuller solutions [3].

Table 7. How the team moved from the research evidence to three technical concepts.

Step	Method used	What the team did	What the discussion produced
Open the frame	Research led mind mapping	The team linked the difficulties found in Part 1 to ways the platform might respond. It then used the design question in Section 3.1 to decide which responses remained relevant.	The team retained only responses that remained within the final problem statement.
Keep alternatives open	Mechanism based divergence	As the team discussed, the initial directions asked the system to verify an option, compare daily fit, guide first contact and make an agreement explicit.	The discussion kept four starting directions visible instead of merging them into one product idea.
Form comparable concepts	Idea bundling	The team grouped guided contact with the trust flow. It kept the matcher focused on explaining daily fit. It formed the agreement concept around how shared expectations become explicit.	Three concepts were ready for the weighted matrix in Section 4.

As shown in the mind map in Appendix A, the early activity opened a broad field and included ideas that later moved outside Sprint 1. As the team discussed, this field was narrowed by keeping separate the mechanism that governs first contact, the one that explains daily fit and the one that records an agreement. The team retained three concepts and delayed selection until the weighted comparison.

4 Concept Selection and Prioritisation

4.1 Selection Method

The weighted decision matrix selected the concept. MoSCoW prioritisation then set the delivery scope. This separation prevented a broad concept from being treated as a commitment to deliver every related feature.

Each concept was rated from 1 to 5. A rating of 1 indicated poor fit. A rating of 3 indicated partial fit with material limits. A rating of 5 indicated full fit under the stated prototype assumptions. The weighted total was calculated as

$$S_j = \sum_i w_i r_{ij}$$

where w_i is the criterion weight and r_{ij} is the rating for concept j.

Table 8. Selection criteria and weights.

Criterion	Weight	Rationale
Research evidence fit	30%	Measures direct traceability to the Part 1 findings and personas.
User risk reduction	20%	Measures how much evidenced user risk the concept reduces.
Feasibility within ten weeks	20%	Measures whether the team can prototype and revise the concept within the schedule.
Prototype testability	15%	Measures whether users can complete observable tasks in a clickable flow.
Innovation beyond isolated tools	10%	Measures whether the concept creates a research grounded journey rather than copying one existing function.
Accessibility and privacy risk	5%	Measures whether foreseeable risks can be controlled. An unacceptable risk remains a rejection condition.

4.2 Decision Matrix

The weighted total is the sum of each rating multiplied by its weight. The maximum score is 5.00.

Table 9. Weighted decision matrix.

Selection criterion	Weight	Concept A	Concept B	Concept C
Research evidence fit	30%	5	5	4
User risk reduction	20%	5	4	4
Feasibility within ten weeks	20%	3	4	5
Prototype testability	15%	4	5	5
Innovation beyond isolated tools	10%	3	4	4
Accessibility and privacy risk	5%	3	4	4
Weighted total	100%	**4.15**	**4.45**	**4.35**

Concept B ranks first with 4.45. Concept C follows at 4.35, so the team retained one limited agreement transition without adopting the full Concept C scope.

Table 10. Evidence for selected decision matrix scores.

Concept	Criterion	Score	Evidence
A	Feasibility within ten weeks	3	The concept requires an identity evidence workflow and protected message handling.
B	Prototype testability	5	Seeded profiles make ranking, filtering and explanations directly observable.
B	Innovation beyond isolated tools	4	The journey connects explainable matching to trust and contact but uses established interaction patterns.
C	Prototype testability	5	Seeded users make proposal, review and approval states directly observable.
C	Research evidence fit	4	Domestic student evidence supports explicit agreements, while accommodation seekers provided weaker support.

4.3 Selected Concept and Rationale

Concept B was selected as the core because it addresses the strongest compatibility evidence and supports an observable comparison task. The final solution is a hybrid. It adds one visible trust state from Concept A and one acknowledgement state from Concept C. Both additions reuse the selected journey and seeded data, so the feasibility rating and Sprint 1 scope remain valid.

The report records product release, sprint assignment and MoSCoW priority as separate fields. Product release records the planned release horizon. Sprint assignment records scheduled work. MoSCoW priority records relative importance.

Table 11. Product release, sprint assignment and MoSCoW priority.

Product release	Sprint assignment	MoSCoW priority	User stories	Delivery decision
MVP	Sprint 1	Must	US-02, US-09, US-10, US-11, US-12, US-13	These stories form one testable path from search to an acknowledged rule.
Post-MVP	Product backlog	Should	US-01, US-03, US-04, US-05, US-06, US-14	These stories extend the agreement into routine household use.
Post-MVP	Product backlog	Could	US-07, US-08, US-15	These stories add optional communication and review functions.
Exploratory	Product backlog	Won't	US-16, US-17, US-18	These stories are outside the current release and test plan.

All MVP stories are assigned to Sprint 1 in the current plan. Product backlog stories are not committed for delivery in this preliminary report. The same three fields are used in the story tables, Story Map and Appendix B.

5 Product Backlog Structure

5.1 Epic Overview

The backlog is organised into four Epics that follow the selected journey. EP-01 contains MVP work. EP-02 contains MVP and Post-MVP work. EP-03 is Post-MVP and EP-04 is exploratory. The Epic grouping does not define release or sprint assignment. Every story retains a stable identifier and maps to one use case in Appendix B.

Table 12. Epic structure of the product backlog.

Epic ID	Epic name	Epic goal and value	Linked user stories
EP-01	Housemate Discovery and Trust	Help accommodation seekers judge whether a listing or a potential housemate can be trusted before any commitment is made.	US-09, US-11, US-12, US-13
EP-02	Shared Household Living	Make household expectations explicit and keep everyday responsibilities visible after move-in.	US-01 to US-08 and US-10
EP-03	Accountability Extensions	Strengthen follow-through on accepted responsibilities once the core journey is validated.	US-14, US-15
EP-04	Property Management	Give property managers a direct channel to publish listings and communicate with their tenants.	US-16, US-17, US-18

5.2 User Stories

Each story uses the standard user, goal and value structure. The three delivery fields follow Table 11.

Table 13. EP-01 Housemate Discovery and Trust user stories.

Story	User story	Release	Sprint	Priority	Jira
US-09	As an international student, I want to message a potential housemate in the app before sharing contact details, so that I can make first contact with less exposure to scams or harassment.	MVP	Sprint 1	Must	ST4-35
US-11	As an international student, I want to see a profile's verification status, so that I can assess the identity signal before contacting that person.	MVP	Sprint 1	Must	ST4-38
US-12	As an accommodation seeker with an irregular schedule, I want to compare potential housemates against my living preferences, so that I can identify practical mismatches before committing.	MVP	Sprint 1	Must	ST4-39
US-13	As a student moving out of home, I want to filter listings by location and cost, so that I can narrow the search to suitable options.	MVP	Sprint 1	Must	ST4-43

Table 14. EP-02 Shared Household Living user stories.

Story	User story	Release	Sprint	Priority	Jira
US-01	As a new share house resident, I want to assign rotating chores and view completion history, so that each contribution remains visible.	Post-MVP	Product backlog	Should	ST4-23
US-02	As an international student in a new household, I want to view the accepted house rules in one place, so that I can understand what the household expects.	MVP	Sprint 1	Must	ST4-24
US-03	As a new share house resident, I want housemates to review a proposed chore system before it becomes active, so that the roster reflects a shared decision.	Post-MVP	Product backlog	Should	ST4-26
US-04	As a share house resident, I want to record a shared expense and who paid it, so that the cost can be divided among the relevant housemates.	Post-MVP	Product backlog	Should	ST4-28
US-05	As a share house resident, I want to view outstanding balances, so that I know who owes money and how much.	Post-MVP	Product backlog	Should	ST4-29
US-06	As an international student, I want to divide a utility bill among selected housemates, so that my share is clear.	Post-MVP	Product backlog	Should	ST4-32
US-07	As a resident who is often away, I want to post household notices in a dedicated space, so that important updates are not lost in chat.	Post-MVP	Product backlog	Could	ST4-33
US-08	As a resident, I want to raise a household issue without showing my identity to other residents, so that I can report a sensitive concern with less social pressure.	Post-MVP	Product backlog	Could	ST4-34
US-10	As a resident with different noise habits from my housemates, I want to propose a house rule and record each acknowledgement, so that the active rule is clear during a dispute.	MVP	Sprint 1	Must	ST4-36

Table 15. EP-03 Accountability Extensions user stories.

Story	User story	Release	Sprint	Priority	Jira
US-14	As a resident, I want an in app reminder when my chore or payment is overdue, so that I can act before a housemate must follow up.	Post-MVP	Product backlog	Should	ST4-25
US-15	As an accommodation seeker, I want to read reviews from confirmed former housemates, so that I can consider past shared living experience before choosing a match.	Post-MVP	Product backlog	Could	ST4-41

Table 16. EP-04 Property Management user stories.

Story	User story	Release	Sprint	Priority	Jira
US-16	As a property manager, I want to record rent status and send overdue reminders, so that I can follow up late payments efficiently.	Exploratory	Product backlog	Won't	ST4-30
US-17	As a property manager, I want to publish an accurate rental listing, so that students can assess the property and enquire through the platform.	Exploratory	Product backlog	Won't	ST4-40
US-18	As a property manager, I want to send an update to every current tenant at a property, so that important information reaches the correct audience.	Exploratory	Product backlog	Won't	ST4-37

6 Acceptance Criteria

The Sprint 1 criteria are linked to prototype evidence in Section 13. The complete Given When Then criteria for every backlog story appear in Appendix C and on the linked Jira tickets.

7 Functional and Non-Functional Requirements

7.1 Verification Access Policy

Phone verification controls messaging access. Identity verification supplies a trust signal and does not unlock discovery or compatibility functions. Table 17 separates these states.

Table 17. Access policy for phone and identity verification.

Account state	Browse listings	View profiles	Compatibility	Send message
Guest or email only	Yes	Yes	Yes	No
Phone verified and identity pending	Yes	Yes	Yes	Yes, with an unverified identity warning
Phone and identity verified	Yes	Yes	Yes	Yes, with a verified identity badge

7.2 Functional Requirements

Table 18. Sprint 1 functional requirements.

ID	Functional requirement	Related stories	Verification method
FR-01	The system shall search from a campus or custom address, apply every active filter and show an affordability comparison. It shall accept reports of misleading listings.	US-13	Prototype AC-US13-01 to 04 and 06. Future system validation AC-US13-05.
FR-02	The system shall record identity review status. Profiles shall show that status, and search shall support a verified only filter.	US-11	Prototype AC-US11-01 is simulated and AC-US11-02 is implemented. Future system validation AC-US11-03 to 06.
FR-03	The system shall rank profiles from stored lifestyle preferences. Each profile shall show the score and explain key differences before contact.	US-12	Prototype AC-US12-01, 02 and 04 to 06. Future system validation AC-US12-03.
FR-04	Phone verified users shall send messages without exposing contact details. The browser shall reject contact details before submission. An identity pending sender shall produce a recipient warning.	US-09	Prototype AC-US09-01 and 03 are simulated, while AC-US09-04 is implemented. Future system validation AC-US09-02, 05 and 06.
FR-05	The system shall show all household rules in one view. It shall use the selected language and fall back to simplified English when translation is unavailable.	US-02	Prototype AC-US02-01 to 03. Future system validation AC-US02-04 to 06.
FR-06	The system shall collect acknowledgement from every housemate before a proposed rule becomes active. Later changes shall require majority approval.	US-10	Prototype AC-US10-01, 02 and 04. Future system validation AC-US10-03, 05 and 06.

7.3 Non-Functional Requirements

Table 19. Non-functional requirements.

ID	Category	Measurable requirement	Verification method
NFR-01	Privacy	Personal contact data shall remain hidden until both users consent. Every role except the verification role shall be denied access to identity evidence.	Run AC-US09-01 and AC-US09-02. Test every role against stored identity evidence.
NFR-02a	Prototype privacy	Contact detail detection shall run in the browser before message submission. A rejected message shall not enter the conversation.	Enter email addresses and phone numbers. Confirm that local validation blocks each message before submission.
NFR-02b	Production security	Message content shall use end-to-end encryption with message keys unavailable to the platform server. Identity evidence shall remain encrypted during transfer and storage.	Inspect the production architecture and run interception and role access tests. This control is outside the preliminary prototype.
NFR-03	Accessibility	Every Sprint 1 screen shall support keyboard use and meet WCAG 2.2 AA contrast thresholds.	Run a keyboard walkthrough and contrast audit on S-01 to S-06.
NFR-04	Performance	Under Chrome 150.0.7871.125 at a 390 by 844 pixel viewport using the DevTools Fast 4G profile with 1.6 Mbps downstream, 750 Kbps upstream and 150 ms RTT, search and compatibility results shall render within 3.0 seconds at p95 across 20 warm cache runs with 50 seeded listings and 100 seeded profiles.	Run one untimed warm up followed by 20 consecutive searches. The requirement passes when p95 is no more than 3.0 seconds.
NFR-05	Usability	The household rules view shall be reachable within two taps from the home screen.	Count taps from the home screen on each supported mobile layout.
NFR-06	Language access	The rules view shall support at least five target languages. It shall show simplified English if translation fails.	Run AC-US02-02 and AC-US02-06 for every supported language.

8 User Story Dependencies

Each user story maps one to one onto a goal-level use case, so the dependency analysis is presented with use case identifiers and applies directly to the corresponding stories. The analysis also covers four supporting use cases together with two seeded data sources, because they enable several stories at once rather than belonging to a single story. The full mapping between stories and use cases is recorded in Appendix B.

8.1 Dependency Table

Table 20. Dependency types used in the analysis.

Type	Meaning
Required	The dependent use case cannot proceed until the prerequisite is satisfied.
Required data	The dependent use case needs data that the prerequisite creates or maintains.
Required data (OR)	Any one of the listed prerequisites supplies the needed data.
Conditional	The dependency applies only in a specific situation, such as showing a verified badge.
Optional	The relationship adds support but is not required for the main success path.

Table 21. Dependencies within EP-02 Shared Household Living.

Dependent use case	Depends on	Type	Reason
UC-01 Chore roster	SUP-02 Household membership	Required	Only members of an active household can view or manage its chore roster.
UC-02 View household rules	SUP-02 Household membership	Required	Rules belong to a specific household and only members can view or manage them.
UC-02 View household rules	UC-10 Agreed house rules	Required data	Rules must be set and agreed before they can be displayed in a member's preferred language.
UC-03 Propose chore system	UC-01 Chore roster	Required data	A structured chore system cannot be proposed or activated without an existing roster to build on.
UC-03 Propose chore system	SUP-02 Household membership	Required	All approving housemates must be identified members of the same household.
UC-04 Log and split expenses	SUP-02 Household membership	Required	Expenses and each member's share must belong to an authorised household.
UC-05 Track balances	UC-04 Shared expenses	Required data	Balances are computed from logged expenses, so at least one expense must exist to show who owes what.
UC-06 Split utility bills	UC-04 Shared expenses	Required	A utility bill is itself a shared expense, so UC-06 reuses the logging and cost-splitting capability of UC-04 rather than any pre-existing expense data.
UC-07 Household notices	SUP-02 Household membership	Required	The noticeboard is scoped to a single household and only members can post or read notices.
UC-08 Anonymous issues	SUP-02 Household membership	Required	The issue must belong to the reporter's household while hiding their identity from other residents.
UC-08 Anonymous issues	UC-10 Agreed house rules	Optional	Existing house rules give issues a shared reference point during disputes, but an issue can be raised without them.

Table 22. Dependencies within EP-01 Housemate Discovery and Trust.

Dependent use case	Depends on	Type	Reason
UC-09 Private in-app messaging	SUP-01 Account and role access	Required	Messaging remains locked until the user signs in and completes phone verification.
UC-09 Private in-app messaging	UC-11 Verified profiles	Optional	Messaging needs phone verification but not identity approval. UC-11 supplies the pending warning and verified badge.
UC-11 Verified profiles	SUP-04 Verification status or DATA-02 Seeded profiles	Required data (OR)	A stored verification status or a seeded sample profile must exist before a badge can be shown.
UC-12 Lifestyle filtering	UC-11 Verified profiles	Optional	Compatibility remains available before identity approval. UC-11 adds identity badges and the verified only filter.
UC-13 Search listings	UC-17 Managed listings or DATA-01 Seeded listings	Required data (OR)	At least one published listing, whether managed or seeded, must exist before a student can search it.

Table 23. Dependencies within EP-03 Accountability Extensions and EP-04 Property Management.

Dependent use case	Depends on	Type	Reason
UC-14 Overdue reminders	UC-01 Chore roster or UC-04 Shared expenses	Required data (OR)	A reminder needs an overdue chore assignment or an overdue expense to reference.
UC-14 Overdue reminders	SUP-03 Push notifications	Required	Automatic overdue reminders are delivered through the notification service.
UC-15 Housemate reviews	SUP-02 Household or tenancy membership	Required	A review may only be submitted after a confirmed tenancy or shared living arrangement has ended.
UC-16 Rent tracking	SUP-02 Tenancy membership	Required	Rent status needs a confirmed relationship between tenant and property, and selecting an applicant is not enough.
UC-16 Rent tracking	SUP-03 Push notifications	Optional	Push notifications deliver due date and overdue rent reminders outside the active app.
UC-17 Manage listings	SUP-04 Verification status	Conditional	Verification is required only before a verified badge is shown, and an unverified or pending listing may still be saved.
UC-18 Tenant announcements	SUP-02 Tenancy membership	Required	The system needs a confirmed current tenant audience for the selected property.
UC-18 Tenant announcements	SUP-03 Push notifications	Optional	Push notifications deliver the announcement outside the active app.

8.2 Dependency Graph

Figure 1 summarises the use case dependencies across the product backlog.

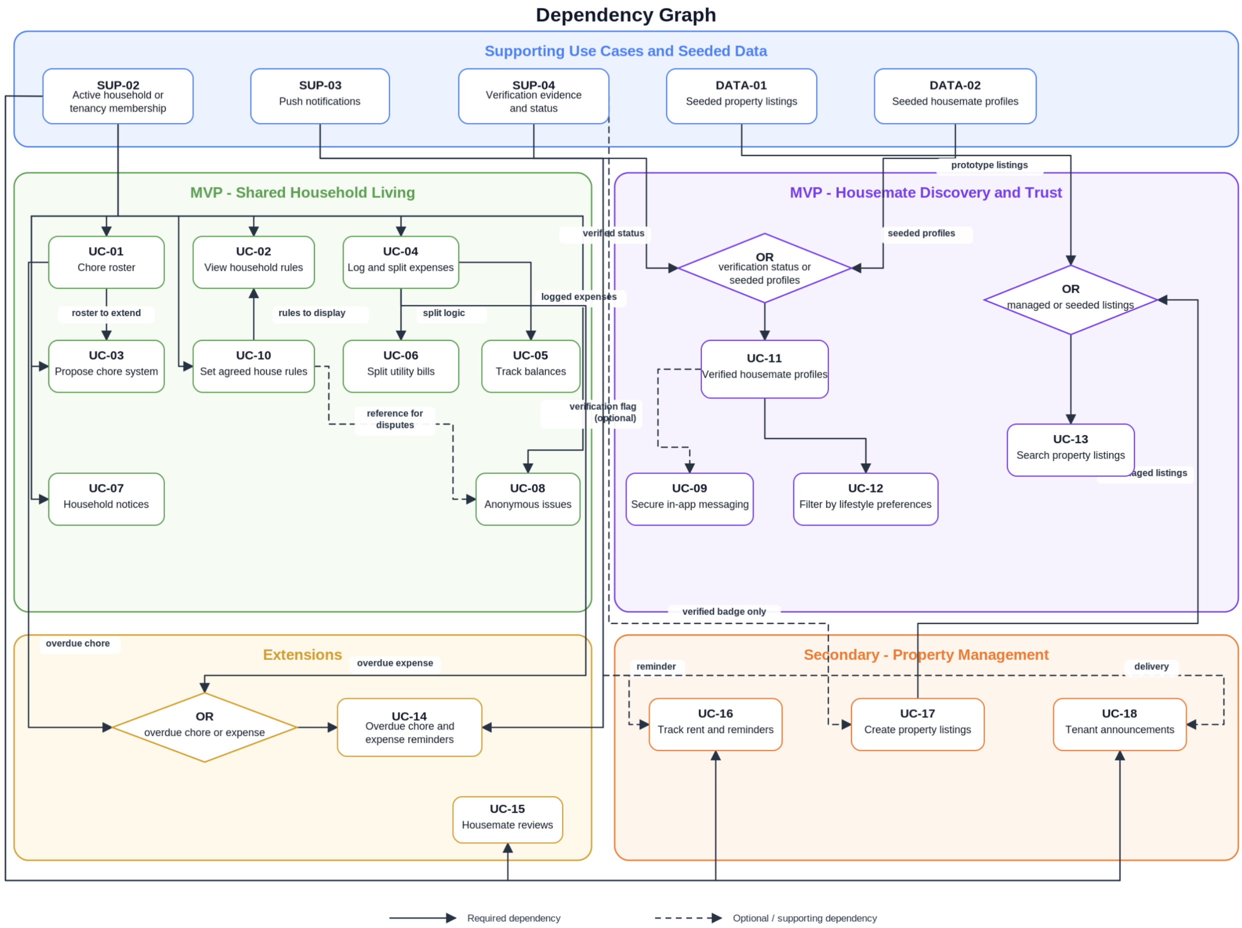


Figure 1. Use case dependency graph for the product backlog.

9 Initial User Story Map

Table 24 applies the same product release, sprint assignment and MoSCoW priority used in the backlog tables. The complete team map remains in Figure 8 in Appendix B.

Table 24. User story map with separate delivery fields.

Release	Sprint	Priority	Journey activity	User task	Stories
MVP	Sprint 1	Must	Find a place	Filter by campus and budget	US-13
MVP	Sprint 1	Must	Evaluate people	Read identity status and lifestyle fit	US-11, US-12
MVP	Sprint 1	Must	Contact safely	Complete phone verification and message in the app	US-09
MVP	Sprint 1	Must	Set expectations	Propose and agree household rules	US-10, US-02
Post-MVP	Product backlog	Should	Manage the shared home	Coordinate chores, bills and reminders	US-01, US-03 to US-06, US-14
Post-MVP	Product backlog	Could	Evaluate people	Read confirmed housemate reviews	US-15
Post-MVP	Product backlog	Could	Manage the shared home	Post notices and raise anonymous issues	US-07, US-08
Exploratory	Product backlog	Won't	Find a place	Create and manage property listings	US-17
Exploratory	Product backlog	Won't	Manage the shared home	Track rent and send tenant announcements	US-16, US-18

9.1 Sprint 1 Scope and Rationale

Sprint 1 contains US-13, US-11, US-12, US-09, US-10 and US-02. Together they form one coherent path. A student narrows affordable properties, evaluates housemates, completes phone verification for private messaging and then accesses explicit household expectations. This sequence tests whether trust information can continue from discovery into the first shared living agreement.

The slice is narrow enough to test with seeded listings and profiles. Post-MVP and exploratory stories remain visible in the product backlog but are not committed to Sprint 1.

10 Preliminary Prototype

10.1 Prototype Scope

The preliminary prototype is an HTML, CSS and JavaScript proof of concept with high visual fidelity. It supports students and young professionals who are searching for shared accommodation. Seeded content allows participants to complete tasks without entering personal data.

Table 25. Prototype scope and fidelity decisions.

Decision	Prototype implementation
Core task path	Search and filter properties (US-13), assess identity status and compatibility (US-11 and US-12), complete phone verification before messaging (US-09), and understand household rules (US-02 and US-10).
Fidelity	Functional high fidelity. Mobile navigation, filters, validation, authentication gates, messaging and multilingual rule interactions are clickable rather than represented by static wireframes.
Visual rationale	Warm photography, warm white surfaces, peach highlights and sage trust signals follow the mood board. Compact layouts and plain labels support repeated use rather than a marketing-style experience.
Prototype data	Seeded listings and profiles provide repeatable test conditions. Changes last for the current page session and reset on refresh, so every participant begins from the same state.
Test environment	The primary assessment viewport is 390 by 844 pixels in Chrome mobile emulation. The responsive desktop layout remains available for secondary review.

This fidelity supports observable interactions linked to user stories without a production backend. Property filtering, compatibility calculation and rule approval run against seeded local data. The phone check uses a demonstration code. Contact detail validation runs in the browser before message submission. Production message encryption and key management are outside the preliminary prototype scope.

10.2 Prototype Flow

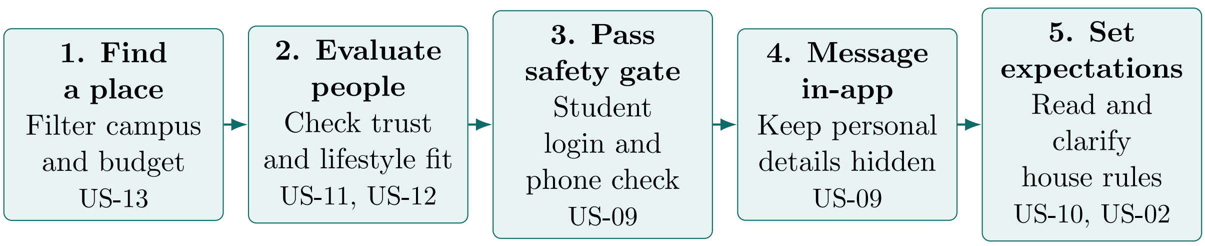
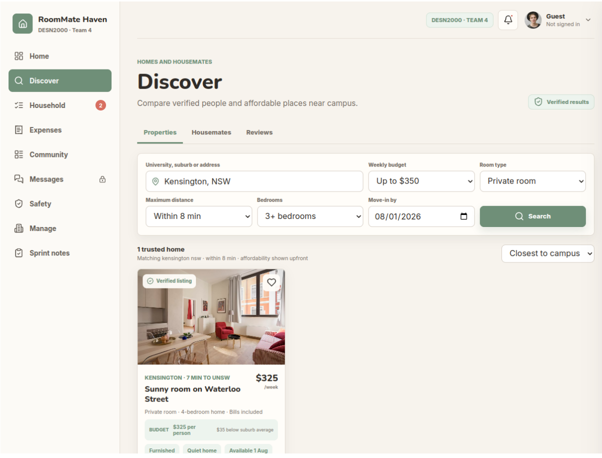
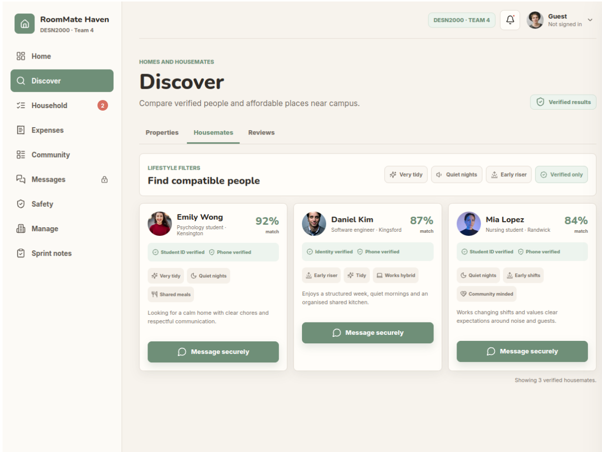


Figure 2. Sprint 1 prototype flow from accommodation search to shared expectations.

10.3 Prototype Screens

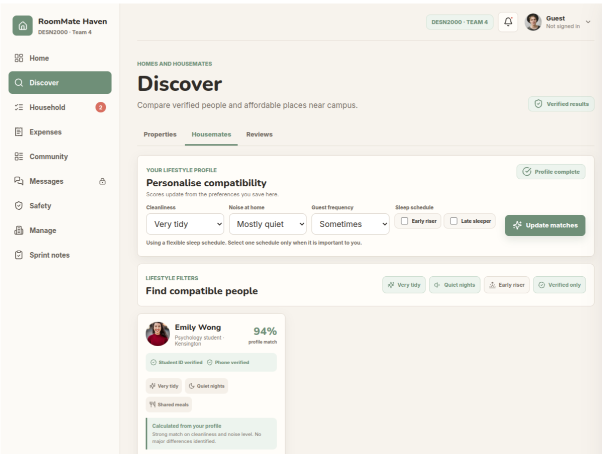


S-01. Property search. Location, budget, room type, distance, bedrooms and move-in date combine into one testable result set.

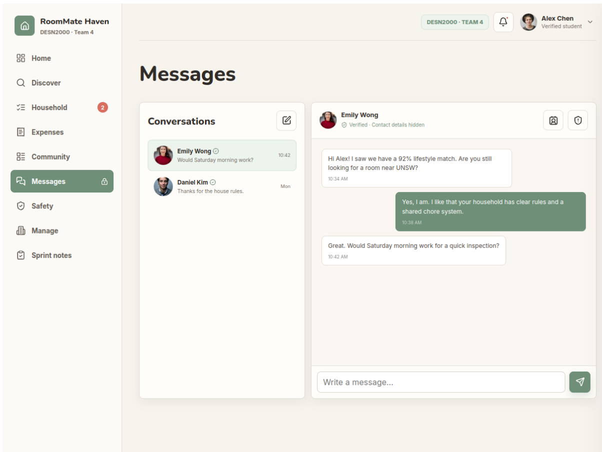


S-02. Verified profiles. Identity and phone badges provide visible trust signals before contact.

Figure 3. Discovery screens for property affordability and profile verification.

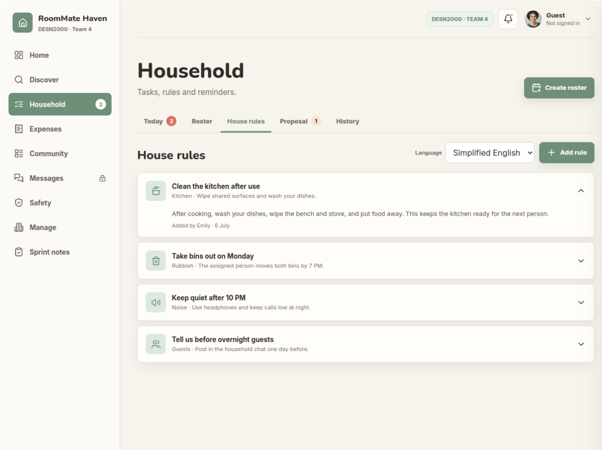


S-03. Lifestyle matching. An editable preference profile recalculates scores, explanations and ranking before filters narrow the results.

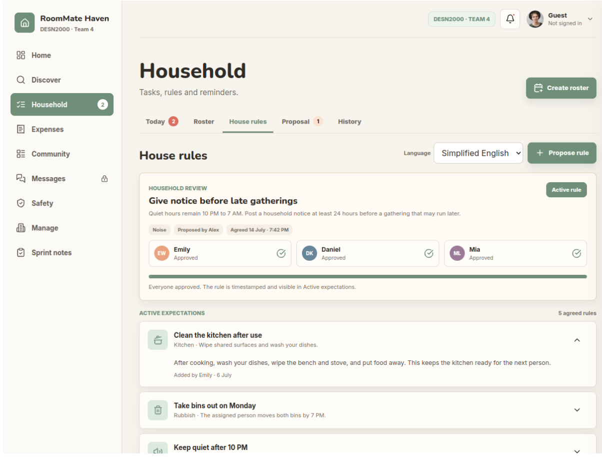


S-04. Private in-app messaging. Contact details stay hidden and messaging is available only after login and phone verification.

Figure 4. Trust and communication screens for comparing and contacting housemates.



S-05. Household rules. All expectations appear in one location with language controls and expandable explanations.



S-06. Unanimous rule approval. A proposal becomes active only after Emily, Daniel and Mia have all approved it.

Figure 5. Shared-expectation screens for accessible and explicit household rules.

These screens expose the feedback required by the Sprint 1 tasks. Section 13 links each interaction to its acceptance criteria.

11 Technical Development Evolution

Figure 6 shows how research narrowed a general accommodation concept into the current Sprint 1 journey.

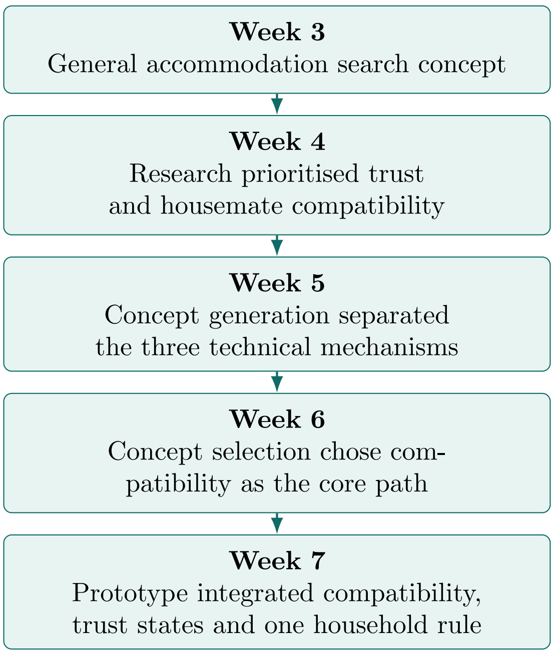


Figure 6. Technical development evolution from Weeks 3 to 7.

12 Innovation Compared to Existing Platforms

Existing accommodation platforms such as Airbnb, Flatmates.com.au and Domain mainly focus on helping users search for available properties and contact landlords or housemates. While these platforms provide useful listing information, they offer limited support for evaluating lifestyle compatibility or preparing users for shared living before moving in.

Our proposed platform goes beyond accommodation search by integrating features specifically designed for university students. It combines lifestyle compatibility matching, verified trust indicators, guided in-app messaging and a simple pre-move-in co-living agreement. These features help users make more informed decisions and reduce potential conflicts before committing to a shared living arrangement.

Rather than replacing existing accommodation platforms, the proposed solution complements them by addressing the trust, compatibility and shared-living challenges identified through our user research. Table 26 summarises how the selected Sprint 1 features compare with three widely used services reviewed during the competitive analysis.

Table 26. Feature comparison between existing accommodation platforms and our proposed platform.

Feature	Airbnb	Flatmates.com.au	Domain	Our Platform
Property listings	✓	✓	✓	✓
Verified listings	Partial	Partial	✓	✓
Lifestyle compatibility matching	–	Limited	–	✓
Guided in-app messaging	Limited	Basic	Basic	✓
Pre-move-in co-living agreement	–	–	–	✓
Student-focused experience	–	Partial	–	✓

Airbnb supports short-term stays and host verification but does not target long-term student housemate matching or pre-move-in household agreements [4]. Flatmates.com.au lists shared accommodation and basic profile filters, yet its compatibility support remains limited and it does not provide a structured agreement workflow before move-in [5]. Domain offers verified rental listings for the Australian market, but its search flow stops at property discovery and does not extend into housemate fit or shared-living preparation [6]. Our platform is the only option in this comparison that combines all six capabilities in one student-focused path.

13 Prototype-to-User-Story Traceability

Each screen maps to the criteria it demonstrates. Table 27 separates implemented interactions from simulated outcomes. Table 28 records every Sprint 1 criterion that remains deferred or production only.

Table 27. Prototype-to-user-story traceability.

ID	Prototype screen	Story	Acceptance criterion	Prototype status	Testable interaction	Link
S-01	Property search	US-13	AC-US13-01 to 04, 06	Implemented	Apply filters, inspect affordability and recover from no results.	Open
S-02	Identity status	US-11	AC-US11-01	Simulated	Inspect a seeded approved identity badge.	Open
S-02	Identity filter	US-11	AC-US11-02	Implemented	Apply the verified only filter.	Open
S-03	Compatibility ranking	US-12	AC-US12-01, 02, 04 to 06	Implemented	Recalculate ranking and test validation and recovery states.	Open
S-04	Private messaging	US-09	AC-US09-01	Simulated	Use a seeded local conversation after phone verification.	Open
S-04	Contact validation	US-09	AC-US09-04	Implemented	Block contact details in the browser before submission.	Open
S-05	Household rule list	US-02	AC-US02-01 to 03	Implemented	Change language and expand a rule explanation.	Open
S-06	Rule approval	US-10	AC-US10-01, 02, 04	Implemented	Validate a proposal and activate it after all approvals.	Open

Table 28. Prototype status for every Sprint 1 acceptance criterion.

Story	Implemented	Simulated	Deferred	Production-only
US-13	01 to 04, 06	–	–	05
US-11	02	01	03, 05	04, 06
US-12	01, 02, 04 to 06	–	03	–
US-09	04	01, 03	02, 06	05
US-02	01 to 03	–	04, 05	06
US-10	01, 02, 04	–	03, 05	06

14 Prototype Test Plan

Table 29. Test overview and coverage of Sprint 1 stories.

Test	Goal	Stories	Acceptance criteria	Responsible
T1	Narrow properties with combined search criteria	US-13	AC-US13-01–04, AC-US13-06	Pranav Shanker
T2	Recognise verified housemate profiles	US-11	AC-US11-01 simulated, AC-US11-02 implemented	Chien-Kai Wang
T3	Build a profile and use recalculated compatibility	US-12	AC-US12-01, AC-US12-02, AC-US12-04–06	Nuodi Liu
T4	Complete authentication and message safely	US-09	AC-US09-01, AC-US09-04	Qian Cao
T5	Propose, review and activate a house rule	US-02, US-10	AC-US02-01–03, AC-US10-01, AC-US10-02, AC-US10-04	Nuodi Liu

Table 30. Test T1. Narrow properties with combined search criteria.

Aspect	Description
Test goal	Verify that a student can combine location, affordability and availability criteria to identify a suitable property.
Related stories	US-13; S-01
Acceptance criteria	AC-US13-01, AC-US13-02, AC-US13-03, AC-US13-04 and AC-US13-06
Target user	Student moving near UNSW with a fixed weekly budget.
Task scenario	Find a private room in Kensington within eight minutes of UNSW, in a household with at least three bedrooms, available by 1 August, for no more than \$350 per week. Explain its affordability rating.
Success criteria	$\geq 4/5$ users enter all six criteria, reach the single matching listing and explain its Budget indicator in ≤ 3 minutes. A deliberately impossible combination must produce the no-result explanation and broadening action.
Expected errors	Users overlook a second-row filter, interpret the move-in date in the wrong direction, or confuse household bedrooms with available rooms.
Data to collect	Completion rate, time on task, filters changed, selected listing, affordability interpretation and recovery from the empty state.
Responsible	Pranav Shanker

Table 31. Test T2. Recognise verified housemate profiles.

Aspect	Description
Test goal	Verify that visible identity signals help users recognise which profiles have been checked.
Related stories	US-11; S-02
Acceptance criteria	AC-US11-01, AC-US11-02
Target user	International student assessing unfamiliar potential housemates.
Task scenario	Find housemates whose identity and phone have been verified. Show the control used to avoid unverified results and explain the evidence used.
Success criteria	$\geq 4/5$ users locate the verified-only control and correctly identify both badge types in ≤ 90 seconds. Participants must explain that the badge is a trust signal, not a guarantee of compatibility or safety.
Expected errors	Users miss the badge, confuse identity and phone verification, or assume a high match score means identity verified.
Data to collect	Badge identification accuracy, explanation quality, time on task and points of visual confusion.
Responsible	Chien-Kai Wang

Table 32. Test T3. Build a profile and use recalculated compatibility.

Aspect	Description
Test goal	Verify that users can create a complete preference profile and understand how it changes compatibility ranking.
Related stories	US-12; S-03
Acceptance criteria	AC-US12-01, AC-US12-02, AC-US12-04, AC-US12-05 and AC-US12-06
Target user	Student who values a tidy home and quiet nights.
Task scenario	Save a very tidy, mostly quiet profile with occasional guests and a flexible schedule. Identify the best match and explain the score. Select both Early riser and Late sleeper, then apply all three strict lifestyle filters.
Success criteria	$\geq 4/5$ users save the complete profile, identify Emily as the 94% top match and cite a profile-derived reason in ≤ 3 minutes. The contradictory schedule must be rejected, and the zero-result state must offer the closest-match recovery action.
Expected errors	Users treat the score as a safety rating, do not notice that ranking changes after saving, or overlook the schedule conflict message.
Data to collect	Profile completion, ranking interpretation, conflict recovery, empty-state recovery, time on task and reasons given.
Responsible	Nuodi Liu

Table 33. Test T4. Complete authentication and message safely.

Aspect	Description
Test goal	Verify that messaging is inaccessible before phone verification and that the browser blocks personal contact details.
Related stories	US-09; S-04
Acceptance criteria	AC-US09-01, AC-US09-04
Target user	International student making first contact with a potential housemate.
Task scenario	Start signed out and open Messages. Sign in with the demo account, verify the phone using code 2468 and send a question about an inspection. Then try to send alex@example.com.
Success criteria	$\geq 4/5$ users complete login and phone verification and send the normal message in ≤ 3 minutes. Messaging must stay locked beforehand, and the email attempt must not appear in the thread while the privacy warning is visible.
Expected errors	Users close the login modal and expect guest messaging, cannot find the phone-check screen, or misread the warning as a successfully sent message.
Data to collect	Completion at each gate, time on task, failed code entries, message outcome and warning comprehension.
Responsible	Qian Cao

Table 34. Test T5. Propose, review and activate a house rule.

Aspect	Description
Test goal	Verify that a household rule remains a proposal until every housemate approves it, while existing expectations remain accessible.
Related stories	US-02, US-10; S-05 and S-06
Acceptance criteria	AC-US02-01, AC-US02-02, AC-US02-03, AC-US10-01, AC-US10-02 and AC-US10-04
Target user	Student living in a shared apartment for the first time and using English as a second language.
Task scenario	Open the quiet hours explanation, then propose a new late gathering rule. Submit it without a category, then complete it. Record one change request, revise it and approve as Emily, Daniel and Mia.
Success criteria	$\geq$4/5 users trigger the required-field message, send the complete proposal, understand that a change request pauses activation, and reach the Active rule state only after the third approval in $\leq$4 minutes. The rule must then appear in Active expectations with all names and a timestamp.
Expected errors	Users confuse a proposal with an active rule, expect one approval to be sufficient, or miss the revise-and-resubmit action after a requested change.
Data to collect	Validation errors noticed, proposal-state comprehension, approval order, time on task and confirmation that the active timestamp was found.
Responsible	Nuodi Liu

After each round the team will compile results into one summary, rank observed issues by severity and feed them into the next prototype iteration. Failed acceptance criteria will be recorded by identifier. The next technical iteration would replace seeded local data with persistent services, run rule approval through independently authenticated housemate accounts, and connect verification to a secure identity provider.

15 Conclusion

The report defines a Sprint 1 journey in which students compare housemates before phone verified contact and then record one agreed household rule. The seeded prototype demonstrates this journey without claiming production identity checks, encryption or persistence.

The next step is to recruit five participants for the moderated tasks. The team will revise any failed acceptance criteria before adding persistent services.

16 References

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[3] IDEO.org. (n.d.). *Bundle Ideas.* Design Kit. https://www.designkit.org/methods/30.html.

[4] Airbnb. (n.d.). *Help Centre: How Airbnb works.* https://www.airbnb.com.au/help/article/362

[5] Flatmates.com.au. (n.d.). *Find a flatmate or room.* https://flatmates.com.au/

[6] Domain. (n.d.). *Rental property search.* https://www.domain.com.au/rent/

A Ideation Mind Map

Figure 7 shows the early mind map used to widen the solution space before the team formed the three technical concepts.

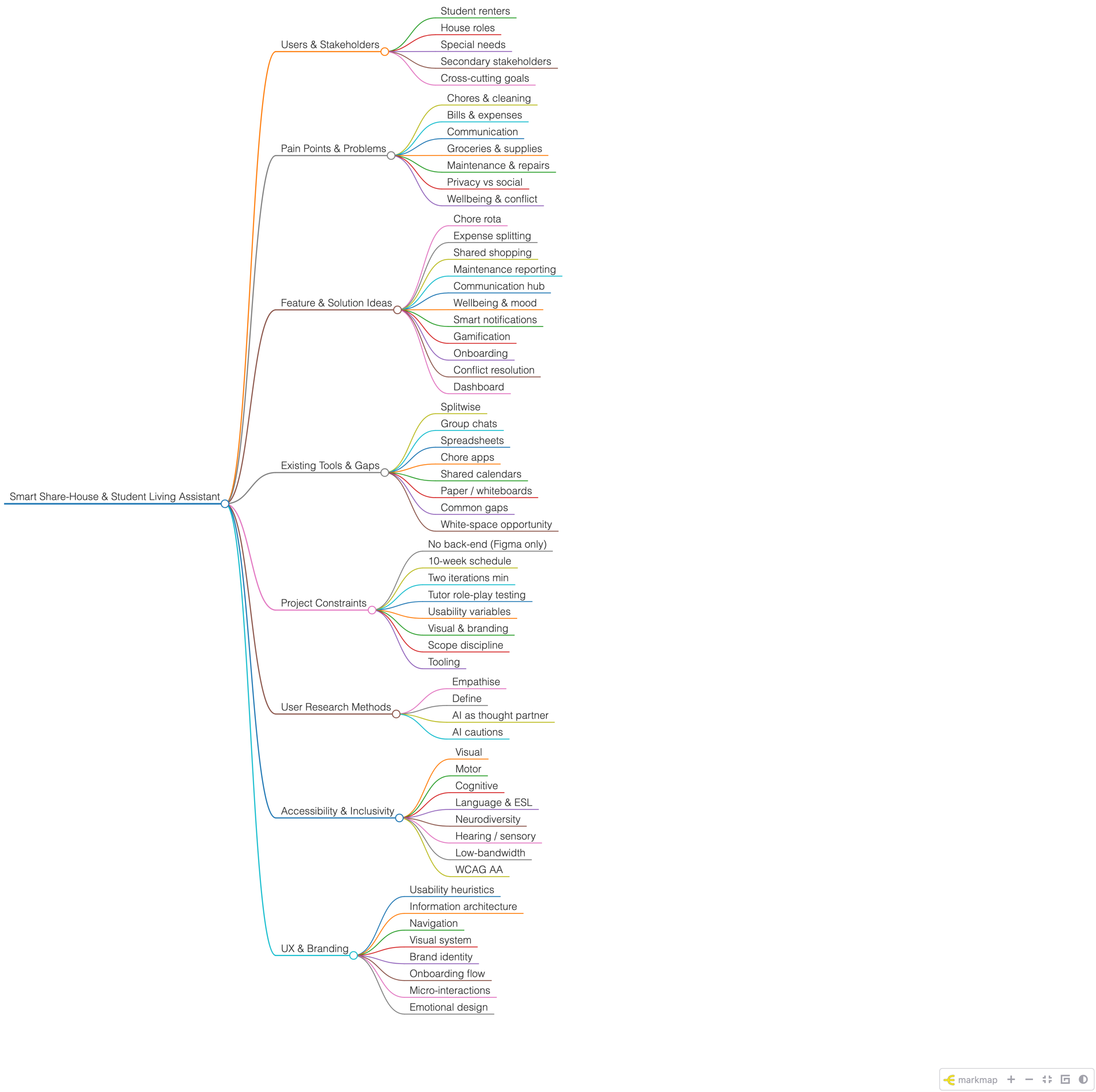

Figure 7. Early mind map used during ideation.

B Full User Story and Use Case Mapping

Figure 8 preserves the complete team map with its personas, activities, backbone tasks and release priorities. Table 24 and the mapping below use the same delivery fields.

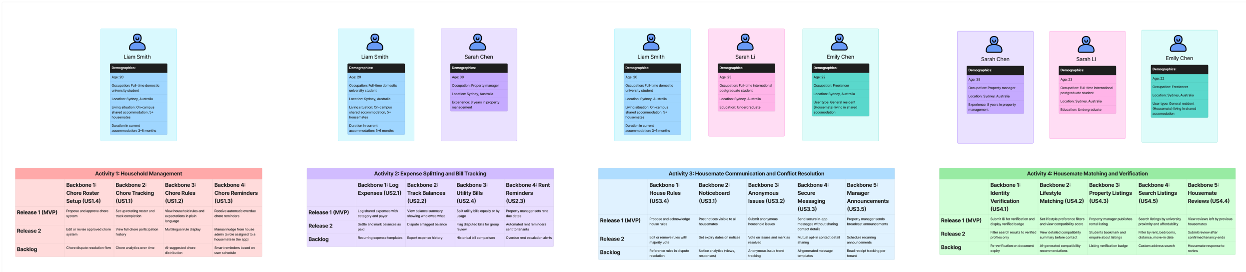

Figure 8. Complete User Story Map showing personas, activities, backbone tasks and release priorities.

Table 35. User story, use case and delivery classification.

Story	Epic	Use case	Release	Sprint	Priority
US-01	EP-02	UC-01 Set up and manage chore roster	Post-MVP	Product backlog	Should
US-02	EP-02	UC-02 View and manage household rules	MVP	Sprint 1	Must
US-03	EP-02	UC-03 Propose and approve chore system	Post-MVP	Product backlog	Should
US-04	EP-02	UC-04 Log and split shared expenses	Post-MVP	Product backlog	Should
US-05	EP-02	UC-05 Track outstanding balances	Post-MVP	Product backlog	Should
US-06	EP-02	UC-06 Split utility bills	Post-MVP	Product backlog	Should
US-07	EP-02	UC-07 Post and view household notices	Post-MVP	Product backlog	Could
US-08	EP-02	UC-08 Raise anonymous household issues	Post-MVP	Product backlog	Could
US-09	EP-01	UC-09 Send private in-app messages	MVP	Sprint 1	Must
US-10	EP-02	UC-10 Set and view agreed house rules	MVP	Sprint 1	Must
US-11	EP-01	UC-11 View identity status	MVP	Sprint 1	Must
US-12	EP-01	UC-12 Filter housemates by lifestyle preferences	MVP	Sprint 1	Must
US-13	EP-01	UC-13 Search and filter property listings	MVP	Sprint 1	Must
US-14	EP-03	UC-14 Receive overdue chore and expense reminders	Post-MVP	Product backlog	Should
US-15	EP-03	UC-15 Read and submit housemate reviews	Post-MVP	Product backlog	Could
US-16	EP-04	UC-16 Track rent and send reminders	Exploratory	Product backlog	Won't
US-17	EP-04	UC-17 Create and manage property listings	Exploratory	Product backlog	Won't
US-18	EP-04	UC-18 Send tenant announcements	Exploratory	Product backlog	Won't

Supporting use cases enable several user goals and do not map to one story. Seeded data lets the prototype demonstrate discovery without implementing the exploratory property workflow.

Table 36. Supporting use cases.

ID	Supporting use case	Primary actor	Successful outcome
SUP-01	Manage account and role access	Any registered user	The user is authenticated and receives the correct student, resident, or property-manager permissions.
SUP-02	Establish active household or tenancy membership	Resident / property manager / selected tenant	Membership is activated only after a household invitation or tenancy is confirmed.
SUP-03	Deliver push notifications	System	Chore reminders, expense reminders, announcement deliveries, and issue notifications can reach users outside the active app.
SUP-04	Manage verification evidence and status	User / property manager / system	Evidence is stored and the related profile receives a pending, verified, or rejected status.

Table 37. Supporting prototype data.

Data ID	Data source	Purpose
DATA-01	Seeded property listings	Provides sample listings while the exploratory property listing workflow remains outside the current release.
DATA-02	Seeded housemate profiles	Provides sample profiles for compatibility and identity badge testing.

C Complete Acceptance Criteria

C.1 EP-01 Housemate Discovery and Trust

Table 38. Acceptance criteria for US-09 (Jira ST4-35).

Acceptance ID	Scenario	Given	When	Then
AC-US09-01	Success	two users are connected on the platform	they open a conversation	they can send and receive messages through a private in-app system without either party's personal contact details being visible
AC-US09-02	Success	both users mutually agree to share contact details	they both opt in	the system exchanges their contact details between them
AC-US09-03	Success	a user feels unsafe	they block or report another user from the messaging interface	the reported user is flagged for review and the conversation is ended
AC-US09-04	Error	a user enters personal contact details in a message	the browser checks the content before submission	it blocks the message and directs the user to the mutual contact sharing feature
AC-US09-05	Error	a user attempts to contact someone who has blocked them	they try to send a message	the system prevents the message from being sent without revealing that they have been blocked
AC-US09-06	Edge case	a phone verified user has not completed identity verification	they initiate a conversation	the recipient sees an unverified identity warning before responding

Table 39. Acceptance criteria for US-11 (Jira ST4-38).

Acceptance ID	Scenario	Given	When	Then
AC-US11-01	Success	a user has submitted a valid form of identification	the verification is approved	their profile displays a verified identity badge visible to all other users
AC-US11-02	Success	a user is searching for housemates	they apply the verified only filter	only profiles with a verified identity badge appear in search results
AC-US11-03	Success	a user views an unverified profile	they open it	a visible warning indicator is displayed clearly at the top of the profile
AC-US11-04	Error	a user submits an expired or unreadable identification document	the system processes it	the verification is rejected and the user is notified with instructions to resubmit a valid document
AC-US11-05	Error	a user has not completed identity verification	they attempt to display a verified identity badge	the system keeps the profile unverified and directs them to complete identity verification
AC-US11-06	Edge case	a verified user's identification document expires	the system detects the expiry	it notifies the user to resubmit updated documentation to maintain their verified status

Table 40. Acceptance criteria for US-12 (Jira ST4-39).

Acceptance ID	Scenario	Given	When	Then
AC-US12-01	Success	a user has completed their lifestyle preference profile	they search for housemates	results are automatically ranked by a compatibility score calculated from matching preferences
AC-US12-02	Success	a user wants to evaluate a potential housemate	they open their profile	a detailed compatibility summary is displayed showing areas of strong match and potential differences
AC-US12-03	Success	a potential housemate views the user's profile	they open it	they receive a notification that someone has viewed their profile
AC-US12-04	Error	a user has not completed their lifestyle preference profile	they attempt to search for compatible housemates	the system prompts them to complete their profile before search results can be personalised
AC-US12-05	Error	a user sets contradictory preferences (e.g. early riser and late sleeper simultaneously)	they attempt to save	the system flags the conflict and prompts them to review their selections
AC-US12-06	Edge case	no housemates match a user's strict preference filters	the search returns zero results	the system suggests broadening one or more filters and displays the closest available matches

Table 41. Acceptance criteria for US-13 (Jira ST4-43).

Acceptance ID	Scenario	Given	When	Then
AC-US13-01	Success	a student enters their university name as a reference point	they search for listings	results are displayed ranked by proximity to the selected campus
AC-US13-02	Success	a student applies filters for maximum weekly rent, distance from campus, number of bedrooms and move-in date	they search	only listings matching all applied filters are displayed
AC-US13-03	Success	a student finds a listing of interest	they open it	they can view affordability indicators including weekly rent in AUD, estimated cost per person, a comparison to the local suburb average and a Budget/Moderate/Premium affordability rating
AC-US13-04	Error	a student searches for listings in an area with no available results	the search returns empty	the system notifies them and suggests broadening their search radius or adjusting filters
AC-US13-05	Error	a listing contains inaccurate photos or descriptions	a student reports it	the listing is flagged for review and temporarily hidden from search results if multiple reports are received
AC-US13-06	Edge case	a student's preferred university is not listed in the system	they search	the system allows them to enter a custom address or suburb as an alternative reference point

C.2 EP-02 Shared Household Living

Table 42. Acceptance criteria for US-01 (Jira ST4-23).

Acceptance ID	Scenario	Given	When	Then
AC-US01-01	Success	a user has created a household and added housemates	they set up a chore roster with task assignments and a rotation schedule	the system distributes chores evenly and notifies each housemate of their assigned tasks
AC-US01-02	Success	a housemate has been assigned a chore	they mark it as complete	the system logs the completion with a timestamp and updates the participation summary
AC-US01-03	Success	a user wants to review contributions	they access the chore history	they can see a full log of who completed what and when
AC-US01-04	Error	a user attempts to create a roster without assigning all housemates	they submit	the system prompts them to assign at least one task to each housemate before activating
AC-US01-05	Error	a chore deadline has passed without being marked complete	the system checks for overdue tasks	it flags the chore as overdue and notifies the assigned housemate
AC-US01-06	Edge case	a housemate leaves the household	their assigned chores become unassigned	the system alerts the house admin to reassign those tasks

Table 43. Acceptance criteria for US-02 (Jira ST4-24).

Acceptance ID	Scenario	Given	When	Then
AC-US02-01	Success	a housemate has added household rules and expectations	an international student opens the household rules section	they can view all rules in a single clearly organised list
AC-US02-02	Success	a user has set their language preference to a non-English language	they view household rules	the rules are displayed in their preferred language or simplified English
AC-US02-03	Success	a user wants clarification on a specific rule	they tap on it	they can view an expanded description or explanation provided by the housemate who created it
AC-US02-04	Error	a user attempts to save a rule without a title or description	they submit	the system prompts them to complete all required fields before saving
AC-US02-05	Error	no household rules have been added yet	a user opens the rules section	the system displays a prompt encouraging housemates to add expectations
AC-US02-06	Edge case	a rule is written in English but the user's preferred language is different	the translation service is unavailable	the system displays the rule in simplified English as a fallback

Table 44. Acceptance criteria for US-03 (Jira ST4-26).

Acceptance ID	Scenario	Given	When	Then
AC-US03-01	Success	a user wants to establish a chore system	they submit a proposed chore system including task assignments and schedules	all housemates receive a notification to review the proposal
AC-US03-02	Success	all housemates have reviewed the proposal	they all approve it	the chore system becomes active and assignments are distributed accordingly
AC-US03-03	Success	a housemate wants to suggest changes	they submit feedback on the proposal	the proposer receives a notification and can revise and resubmit
AC-US03-04	Error	a housemate does not respond to the proposal within 7 days	the deadline passes	the system sends a reminder and flags the proposal as awaiting response
AC-US03-05	Error	a user submits a proposal without assigning tasks to all housemates	they attempt to submit	the system prompts them to assign at least one task per housemate
AC-US03-06	Edge case	one housemate rejects the proposal	the rejection is submitted	the system notifies the proposer and pauses activation until a revised proposal is approved by all

Table 45. Acceptance criteria for US-04 (Jira ST4-28).

Acceptance ID	Scenario	Given	When	Then
AC-US04-01	Success	a user has paid for a shared expense	they log it with a description, amount, category and payer	the system automatically calculates each selected housemate's share and notifies them
AC-US04-02	Success	an expense has been logged	a housemate views their dashboard	they can see all expenses they are involved in and their individual amount owed
AC-US04-03	Success	a housemate has paid their share	the payer marks it as settled	the balance between those two housemates updates automatically
AC-US04-04	Error	a user submits an expense without entering an amount	they attempt to save	the system prompts them to complete all required fields
AC-US04-05	Error	a user attempts to split an expense among housemates not in their household	they submit	the system only allows splitting among confirmed household members
AC-US04-06	Edge case	a housemate leaves the household with an outstanding balance	their account is removed	the system retains their balance record and flags it as unresolved for the remaining housemates

Table 46. Acceptance criteria for US-05 (Jira ST4-29).

Acceptance ID	Scenario	Given	When	Then
AC-US05-01	Success	expenses have been logged between housemates	a user opens their balance dashboard	they can see a clear summary of who owes them money and how much
AC-US05-02	Success	a housemate has paid their share	the user marks the balance as settled	the dashboard updates automatically to reflect the new balance
AC-US05-03	Success	balances exist across multiple housemates	the user views the dashboard	all balances are displayed individually per housemate
AC-US05-04	Error	a user attempts to mark a balance as settled that has already been settled	they submit	the system notifies them that the balance is already cleared
AC-US05-05	Error	no expenses have been logged yet	a user opens the balance dashboard	the system displays a prompt to add the first shared expense
AC-US05-06	Edge case	a balance is disputed by a housemate	they flag it	the system marks the balance as disputed and notifies both parties to resolve it before it can be settled

Table 47. Acceptance criteria for US-06 (Jira ST4-32).

Acceptance ID	Scenario	Given	When	Then
AC-US06-01	Success	a utility bill has been received	a user enters the bill type, total amount and billing period	the system calculates and assigns each housemate's share based on the selected split method
AC-US06-02	Success	a bill has been logged	all housemates view the shared bill history	they can see all submitted bills and their individual amounts owed
AC-US06-03	Success	a housemate believes a bill is incorrect	they flag it as disputed	all housemates are notified and the bill is marked as under review
AC-US06-04	Error	a user submits a bill without entering a total amount	they attempt to save	the system prompts them to complete all required fields
AC-US06-05	Error	a user attempts to split a bill among zero housemates	they submit	the system requires at least two participants before allowing submission
AC-US06-06	Edge case	a utility bill covers a period during which a housemate was absent	the bill is split	the system allows the user to manually exclude or adjust that housemate's share for that billing period

Table 48. Acceptance criteria for US-07 (Jira ST4-33).

Acceptance ID	Scenario	Given	When	Then
AC-US07-01	Success	a user wants to post a notice	they create a notice with a title, description and optional image	it is immediately visible to all housemates on the noticeboard
AC-US07-02	Success	a notice has been posted	a housemate wants to respond	they can leave a limited reaction or reply directly on the notice
AC-US07-03	Success	a notice has an expiry date set	the date passes	the notice is automatically archived and removed from the active noticeboard
AC-US07-04	Error	a user attempts to post a notice without a title	they submit	the system prompts them to add a title before posting
AC-US07-05	Error	a user sets an expiry date in the past	they submit	the system rejects the date and prompts them to select a future date
AC-US07-06	Edge case	a notice receives no interaction for 30 days and has no expiry date set	the system checks for inactive notices	it prompts the creator to either renew or archive the notice

Table 49. Acceptance criteria for US-08 (Jira ST4-34).

Acceptance ID	Scenario	Given	When	Then
AC-US08-01	Success	a user wants to raise an issue	they submit it with the anonymous option selected	the issue is posted to the shared household space without revealing the submitter's name or profile
AC-US08-02	Success	an issue has been posted	housemates view it	they can vote on whether they agree it is a concern
AC-US08-03	Success	all parties have agreed the issue is resolved	the submitter or a majority vote marks it as resolved	the issue is closed and archived
AC-US08-04	Error	a user attempts to submit an issue without a description	they submit	the system prompts them to add a description before posting
AC-US08-05	Error	a user attempts to identify an anonymous submitter through the system	they try to access submitter information	the system does not reveal any identifying information under any circumstance
AC-US08-06	Edge case	an anonymous issue receives no votes or responses within 14 days	the system checks for inactive issues	it sends a reminder to all housemates to review and respond

Table 50. Acceptance criteria for US-10 (Jira ST4-36).

Acceptance ID	Scenario	Given	When	Then
AC-US10-01	Success	a user wants to propose a house rule	they submit it with a title, description and category	all housemates receive a notification to review and acknowledge it
AC-US10-02	Success	all housemates have acknowledged a proposed rule	the last acknowledgement is submitted	the rule is officially added to the household rules list with a timestamp
AC-US10-03	Success	a user wants to reference a rule during a dispute	they link it to a household issue	the rule is displayed alongside the issue for all housemates to see
AC-US10-04	Error	a user attempts to submit a rule without a title or category	they submit	the system prompts them to complete all required fields
AC-US10-05	Error	a housemate attempts to edit or remove a rule without majority approval	they submit the change	the system rejects it and notifies all housemates that a change has been requested pending approval
AC-US10-06	Edge case	a rule has not been acknowledged by one housemate after 7 days	the system checks for pending acknowledgements	it sends a reminder to that housemate to review and acknowledge the rule

C.3 EP-03 Accountability Extensions

Table 51. Acceptance criteria for US-14 (Jira ST4-25).

Acceptance ID	Scenario	Given	When	Then
AC-US14-01	Success	a chore deadline has passed and the chore has not been marked complete	the system checks for overdue tasks	it automatically sends an in-app notification to the assigned housemate
AC-US14-02	Success	the house admin wants to manually remind a housemate	they send a nudge	the assigned housemate receives an in-app notification without any personal contact details being exchanged
AC-US14-03	Success	a housemate marks an overdue chore as complete after receiving a reminder	they submit the completion	the system updates the chore log and removes the overdue flag
AC-US14-04	Error	a housemate has disabled notifications	a chore becomes overdue	the system still flags the chore as overdue within the app and displays it prominently on their dashboard
AC-US14-05	Edge case	a housemate is temporarily removed from the roster (e.g. on leave)	their chore deadline passes	the system does not send them an overdue reminder and reassigns the task to the next available housemate

Table 52. Acceptance criteria for US-15 (Jira ST4-41).

Acceptance ID	Scenario	Given	When	Then
AC-US15-01	Success	a confirmed tenancy period has ended	a user submits a written review and star rating for their previous housemate	the review is displayed on that housemate's profile in chronological order
AC-US15-02	Success	a user views a housemate's profile	they scroll to the reviews section	they can see all reviews with the overall star rating displayed prominently at the top
AC-US15-03	Success	a housemate receives a review	it is posted	they receive a notification and can submit a single written response
AC-US15-04	Error	a user attempts to submit a review before a confirmed tenancy period has ended	they try to submit	the system prevents submission and notifies them that reviews can only be left after a confirmed tenancy
AC-US15-05	Error	a user attempts to submit a second review for the same housemate from the same tenancy	they submit	the system rejects the duplicate and notifies the user
AC-US15-06	Edge case	a review is flagged as inappropriate by multiple users	the system detects multiple reports	the review is temporarily hidden pending platform review and the reviewer is notified

C.4 EP-04 Property Management

Table 53. Acceptance criteria for US-16 (Jira ST4-30).

Acceptance ID	Scenario	Given	When	Then
AC-US16-01	Success	a property manager has set a rent amount and due date for a tenant	the due date approaches	the system automatically sends a rent reminder to that tenant
AC-US16-02	Success	a tenant's payment has been recorded	the property manager views the dashboard	the tenant's status updates to paid
AC-US16-03	Success	the due date has passed without payment being recorded	the system checks payment status	the tenant's status updates to overdue and the property manager is notified
AC-US16-04	Error	a property manager attempts to set a rent due date without entering a rent amount	they submit	the system prompts them to complete all required fields
AC-US16-05	Error	a tenant's contact details are incomplete	the system attempts to send a reminder	the system flags the tenant's profile as incomplete and alerts the property manager
AC-US16-06	Edge case	a tenant disputes a rent amount	they flag it through the app	the system marks the payment status as disputed and notifies the property manager to review

Table 54. Acceptance criteria for US-17 (Jira ST4-40).

Acceptance ID	Scenario	Given	When	Then
AC-US17-01	Success	a property manager has completed a listing with photos, description, rent price, available date and house rules	they publish it	it immediately appears in student search results based on location and filters
AC-US17-02	Success	a student finds a listing they are interested in	they send an enquiry	the property manager receives a notification and the message is stored in their centralised inbox
AC-US17-03	Success	a listing's available date has passed	the system checks active listings	the listing is automatically removed from search results and the property manager is notified
AC-US17-04	Error	a property manager attempts to publish a listing without photos or a rent price	they submit	the system prompts them to complete all required fields before publishing
AC-US17-05	Error	a student attempts to enquire about a listing that has already been filled	they submit the enquiry	the system notifies them that the listing is no longer available
AC-US17-06	Edge case	a listing has been live for 60 days without being filled	the system checks listing age	it prompts the property manager to review and update the listing details

Table 55. Acceptance criteria for US-18 (Jira ST4-37).

Acceptance ID	Scenario	Given	When	Then
AC-US18-01	Success	a property manager wants to send an announcement	they compose it with a title, description, category and optional attachment	it is delivered to all tenants in the selected property
AC-US18-02	Success	an announcement has been sent	a tenant opens the app	they receive a push notification and can view the announcement in their notification centre
AC-US18-03	Success	a property manager wants to schedule a recurring announcement	they set a recurrence schedule	the system automatically sends the announcement at the defined intervals
AC-US18-04	Error	a property manager attempts to send an announcement without selecting a property	they submit	the system prompts them to select at least one property before sending
AC-US18-05	Error	a scheduled announcement fails to send	the system detects the failure	it retries delivery and notifies the property manager if the issue persists
AC-US18-06	Edge case	a tenant has not viewed a high priority announcement within 24 hours	the system checks read receipts	it sends a follow-up push notification to that tenant

D Prototype Screens and Links

Table 56. Prototype access information.

Item	Access details
Hosted prototype	RoomMate Haven on AWS Elastic Beanstalk
Demo account	Student email `z5555555@ad.unsw.edu.au`. Password `studentdemo`.
Phone verification	Select Send code and enter the demonstration code `2468`.
Reset behaviour	Refresh the page to restore all seeded data and return the prototype to a repeatable test state.
Screenshot evidence	GitHub screenshot folder containing US-01 to US-18 evidence

Table 57. Direct evidence states for Sprint 1 user stories.

Story	Screenshot file	Hosted evidence state
US-02	`US-02_household-rules.png`	Open capture state
US-09	`US-09_secure-in-app-messaging.png`	Open capture state
US-10	`US-10_agreed-noise-rules.png`	Open capture state
US-11	`US-11_verified-housemate-profiles.png`	Open capture state
US-12	`US-12_lifestyle-housemate-filters.png`	Open capture state
US-13	`US-13_property-search-filters.png`	Open capture state

E Jira Evidence

The project space provides Jira and Confluence evidence for the backlog.

Table 58. Atlassian project evidence access.

Evidence item	Access link
Jira and Confluence project space	Open the Atlassian project evidence space